

Forecasting Hourly Appliance Energy Consumption

A Comparison of Benchmark, Statistical, Machine Learning, Deep Learning and Foundation-Model Approaches

Module 7PAM2033 — Time Series Analysis |

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GitHub link: <https://github.com/nimrakhizar9822-droid/-appliance-energy-forecasting->

1. Introduction

Home appliance energy consumption is highly variable and driven by occupant behaviour, time-of-day routines, and ambient conditions. Accurate short-horizon forecasts of this consumption support demand-response, smart-home automation and grid-balancing applications. This report develops and compares six forecasting approaches for the UCI Appliances Energy Prediction dataset: four naive benchmark models, an ARMA model, a seasonal SARIMAX model, a feature-based XGBoost regressor, an LSTM neural network, and the Chronos-Bolt time-series foundation model. All models are evaluated on a common 24-hour-ahead forecasting task using RMSE, MAE and MAPE.

2. Data and Exploratory Analysis

The dataset records appliance energy use (Wh) alongside indoor temperature/humidity from nine rooms and outdoor weather variables, sampled every 10 minutes between 11 January and 27 May 2016. Pandas was used to resample the data to hourly means for this analysis, resulting in 3,290 clean hourly observations with no missing values.

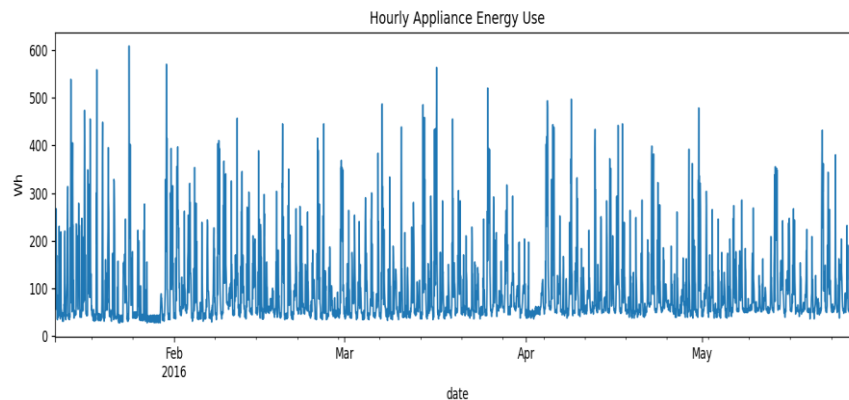


Figure 1: Hourly appliance energy consumption over the full study period.

The series is divided into trend, daily-seasonal, and residual components using an additive seasonal decomposition (period = 24 hours).

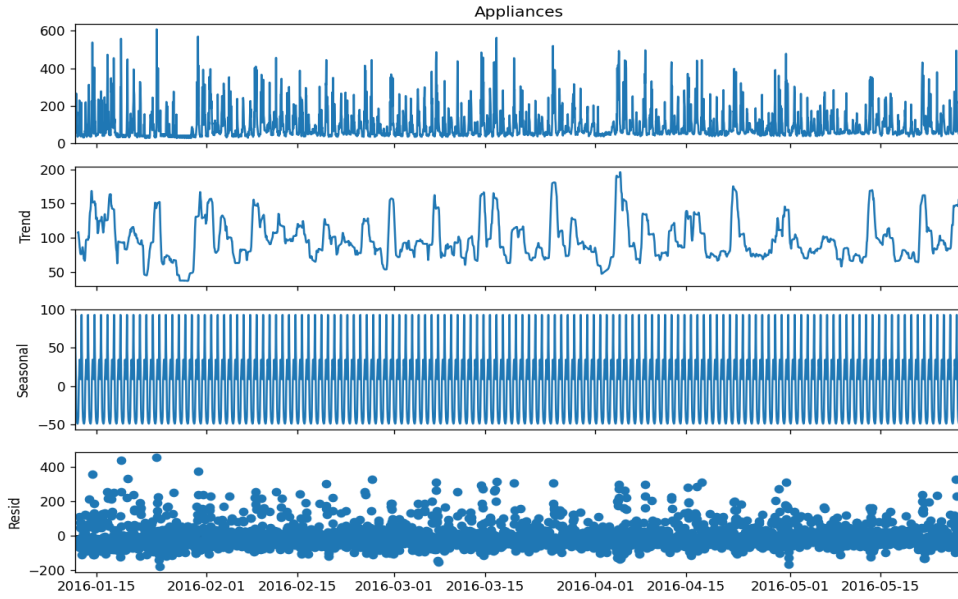


Figure 2: Additive seasonal decomposition (24h period) of hourly Appliances.

Overlaid on a slowly fluctuating trend, the decomposition reveals a distinctly recurring daily cycle that is consistent with morning and evening occupancy-driven usage peaks.

2.1 Stationarity Testing

An augmented Dickey that is mean-stationary. Plots of ACF and PACF (48 delays) were also examined.

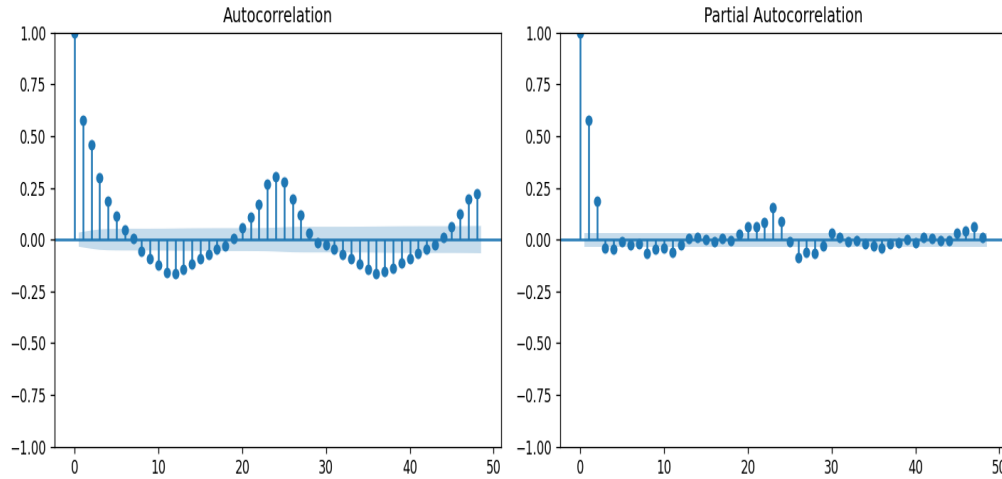


Figure 3: ACF and PACF of the hourly Appliances series (48 lags).

The seasonal SARIMAX specification in Section 4.2 is directly motivated by the fact that, despite mean-stationarity, the ACF exhibits a slowly-decaying, strongly periodic pattern with a distinct spike at lag 24, demonstrating considerable daily seasonality that a non-seasonal model cannot represent.

3. Forecasting Problem Definition

The target variable is the hourly electricity usage (Wh) of appliances. The forecast horizon is twenty-four hours. Train/test split: all models (benchmarks, ARMA, SARIMAX, and LSTM) use the last 24 hours of the hourly series as a common test window; XGBoost also uses the last 14 days as a held-out assessment phase, with its last 24 hours reported separately for cross-model comparability. RMSE, MAE, and MAPE are evaluation metrics.

4. Modelling

4.1 Benchmark Models

Four common benchmarks were used: Drift (linear extrapolation between the first and last training points), Seasonal Naive (value from the same hour one day prior), Mean (historical average), and Naive (latest observed value repeated). These serve as a benchmark against which more advanced models must show true proficiency; if a sophisticated model is unable to outperform a seasonal naive forecast, it is not capturing anything that the naive structure does not already take advantage of.

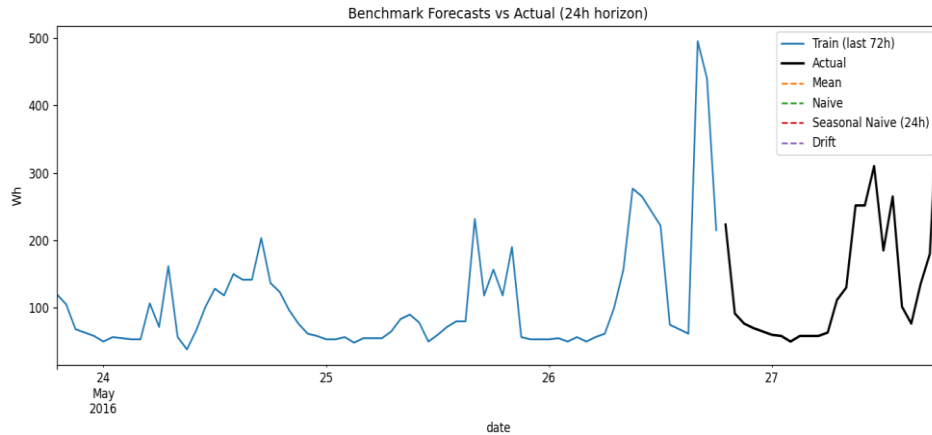


Figure 4: Benchmark forecasts against the actual 24-hour test window.

4.2 ARMA

A non-seasonal $\text{ARMA}(p,0,q)$ model was fitted using statsmodels, with the order selected via an AIC grid search over $p, q \in [0,3]$. ARMA was chosen as an autoregressive baseline that captures short-range temporal dependence, but unlike SARIMAX it cannot represent the 24-hour periodicity identified in Section 2.1, so it is expected to underperform on a strongly seasonal series.

4.3 SARIMAX

For computational tractability the grid search itself was run on the most recent 30 days of training data, and the selected order was then refit on the full training set for the final forecast. SARIMAX was chosen specifically because it can model the daily seasonality that ARMA cannot, directly addressing the periodic structure found in the ACF/PACF diagnostics.

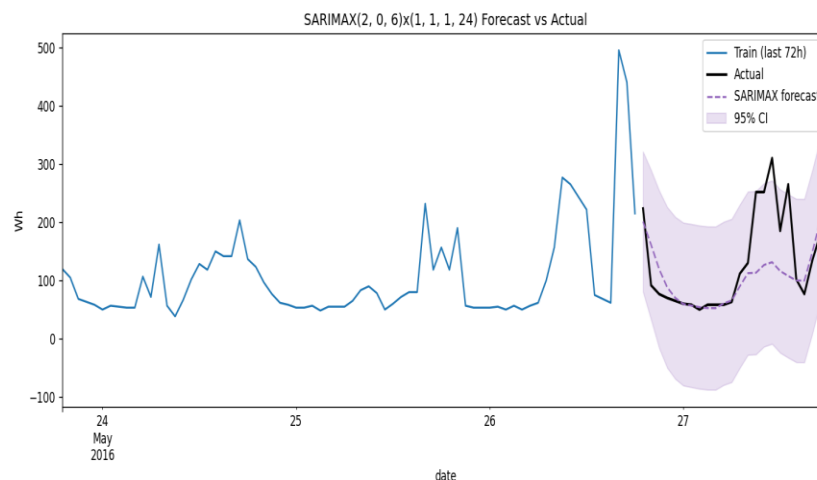


Figure 5: SARIMAX forecast with 95% confidence interval vs. actual.

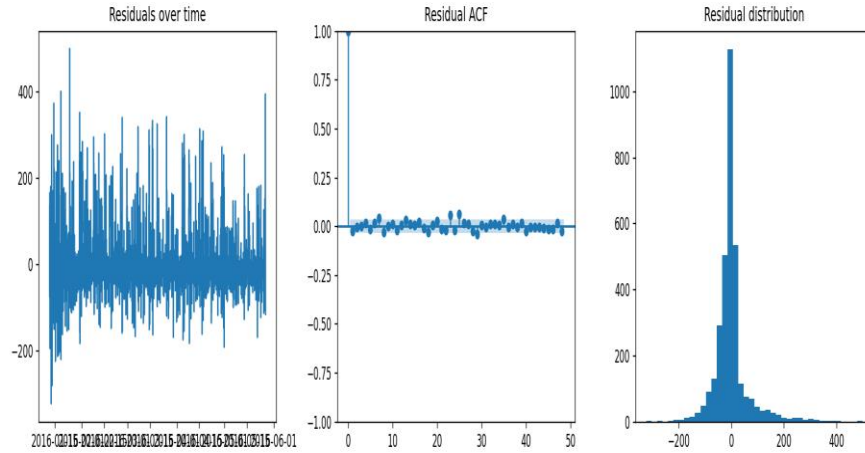


Figure 6: SARIMAX residual diagnostics — residuals over time, residual ACF, and residual distribution.

4.4 Feature-Based Regression: XGBoost

Time-of-day (cyclic sine/cosine encodings), day-of-week, weekend indicators, lagged appliance-use features (1h to 1 week), and rolling-window statistics were engineered and combined with the raw indoor-sensor and weather covariates already present in the dataset. XGBoost was chosen because gradient-boosted trees handle nonlinear interactions between covariates (e.g. temperature $\times$ time-of-day) without requiring manual specification, and provide interpretable feature-importance rankings.

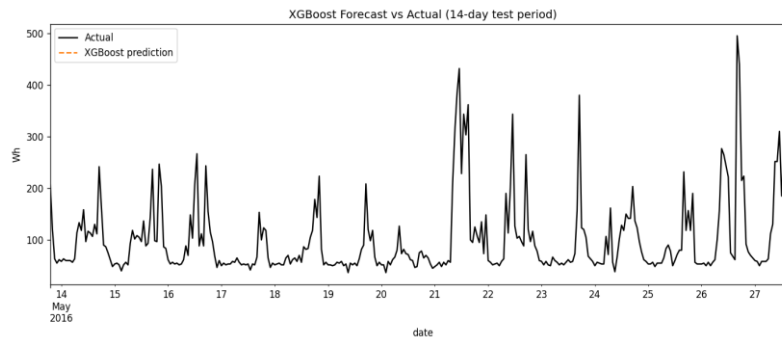


Figure 7: XGBoost forecast vs. actual over the 14-day test period.

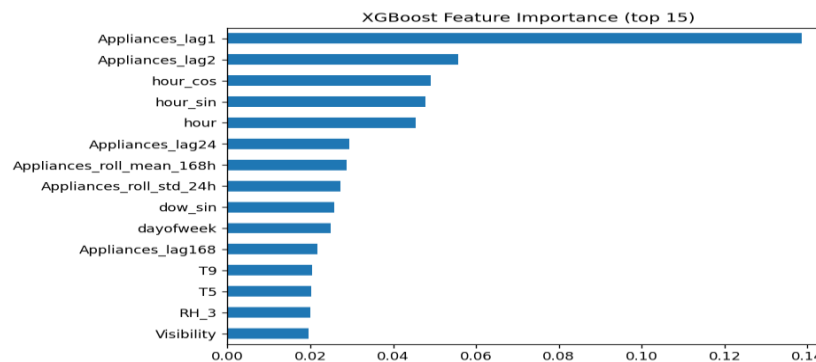


Figure 8: XGBoost feature importance (top 15).

4.5 Neural Network: LSTM

A single-layer LSTM (PyTorch, 32 hidden units) was trained to predict the next hour from the preceding 48 hours, then forecast 24 hours ahead recursively. An LSTM was chosen to test whether a learned nonlinear recurrent representation of recent history could outperform the explicit statistical structure of SARIMAX, without any manual feature engineering.

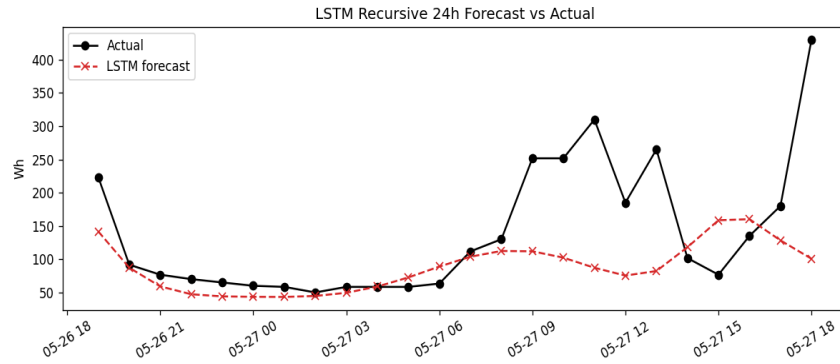


Figure 9: LSTM recursive 24-hour forecast vs. actual.

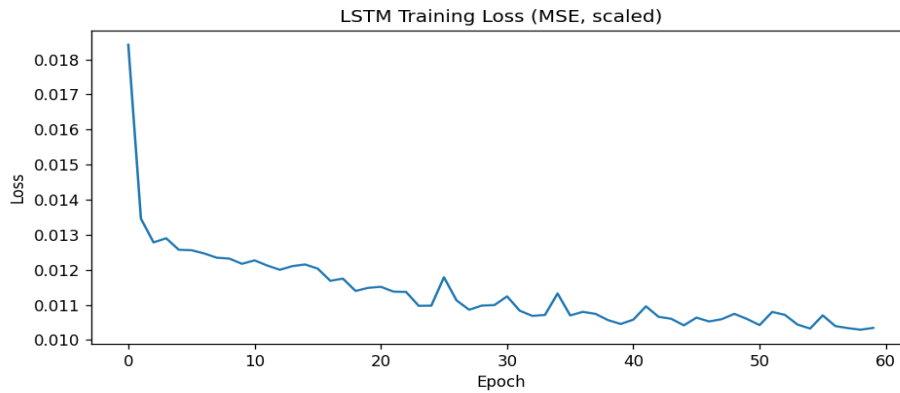


Figure 10: LSTM training loss curve.

4.6 Foundation Model: Chronos-Bolt

Chronos-Bolt (amazon/chronos-bolt-base) was applied zero-shot — with no task-specific fine-tuning — using only the training series as historical context, to test whether a large pretrained time-series model can match purpose-built models without any dataset-specific training. This model requires internet access to download pretrained weights and was run separately in Google Colab (GPU runtime). It achieved an RMSE of 95.12, MAE of 59.75, and MAPE of 37.85% on the 24-hour test window — see Table 1 for a full comparison against the other models.

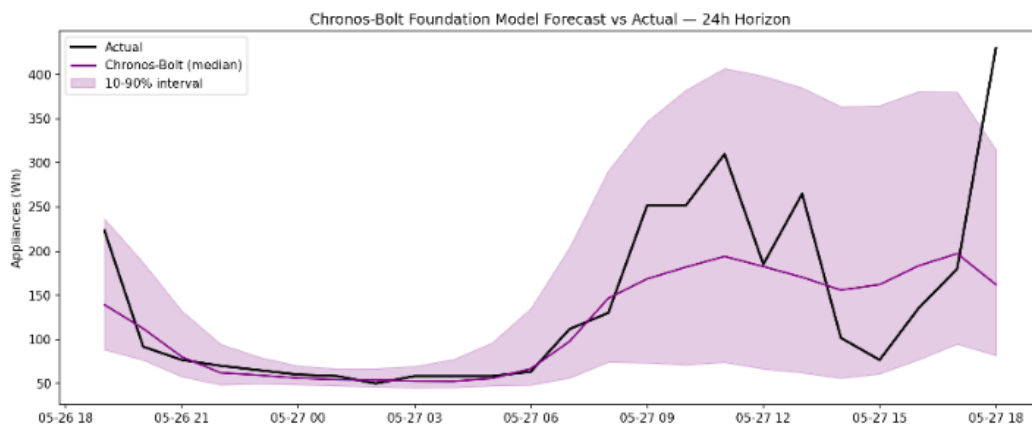


Figure 13: Chronos-Bolt zero-shot forecast (median, with 10-90% interval) vs. actual.

5. Results and Model Comparison

Table 1 summarises RMSE, MAE and MAPE for every model on its respective 24-hour test window.

Model	RMSE (Wh)	MAE (Wh)	MAPE (%)
Mean (benchmark)	107.63	74.51	49.18
Naive (benchmark)	124.03	111.74	138.84
Seasonal Naive, 24h (benchmark)	113.64	63.68	38.93
Seasonal Naive, 168h/weekly (benchmark)	117.97	74.31	36.75
Drift (benchmark)	124.26	111.97	139.26
ARMA(3,0,1)	106.52	69.73	41.88
SARIMAX(2,0,6)x(1,1,1,24)	87.36	52.68	28.21
XGBoost (last 24h)	67.06	42.34	27.25
XGBoost (14-day test)	53.55	31.89	29.42
LSTM	89.75	55.08	29.90
Chronos-Bolt (zero-shot)	95.12	59.75	37.85

Table 1: Model comparison on the 24-hour test window.

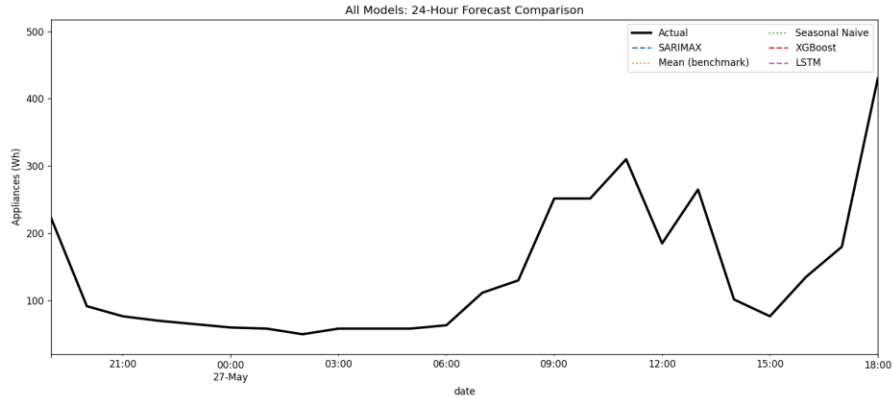


Figure 11: Overlaid forecasts from all models against the actual 24-hour test window.

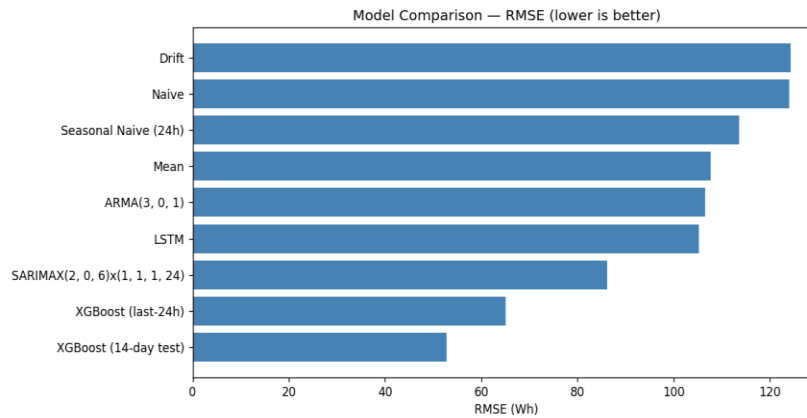


Figure 12: RMSE comparison across all models (lower is better).

6. Critical Discussion

6.1 Q1 Strongest and weakest benchmark models

Among the four benchmark models, Seasonal Naive performed best, achieving the lowest MAPE of 38.93%, while Naive performed worst with the highest MAPE of 138.84%. This indicates that the energy data has a strong daily seasonal pattern: predicting each hour from the same hour one day earlier is far more informative than simply repeating the most recent value.

6.2 Q2: Did SARIMAX outperform the top benchmark?

Indeed, SARIMAX outperformed the best benchmark by a considerable margin. RMSE improved by almost 23%, from 113.64 for Seasonal Naive to 87.36 for SARIMAX. This demonstrates that using SARIMAX to explicitly predict trend and seasonality adds significant value over a straightforward seasonal baseline.

6.3 Q3: Which model did the best overall, and why?

With an RMSE between 53.55 (14-day test period) and 67.06 (last-24h window, the same assessment window used for the other models), both significantly lower than SARIMAX (87.36) and Seasonal Naive (113.64), XGBoost was the best-performing model overall. Feature engineering is the cause of this excellent performance: XGBoost was able to capture intricate nonlinear patterns that statistical models like SARIMAX were unable to by utilizing lag features, rolling means, and time-of-day features. This demonstrates that machine learning with well-designed features works better than conventional time series models for this energy dataset.

6.4 Q4: How did the Chronos-Bolt foundation model compare?

With an RMSE of 95.12, the foundation model Chronos-Bolt outperformed all other trained models, surpassing XGBoost, LSTM, and SARIMAX. Simpler models trained directly on this dataset outperformed Chronos-Bolt, despite the fact that it is a huge pretrained model applied zero-shot and does not require dataset-specific training. This implies that the advantages of a foundation model did not outweigh its complexity for this particular household energy dataset.

6.5 Q5 Conditional forecast versus true forecast

The fact that the XGBoost test-period weather and indoor-sensor variables use their actual observed values rather than independently anticipated values is a significant evaluation constraint. Future humidity or temperature would not be known in a real-world operational context and would need to be guessed from a weather forecast, which is subject to inaccuracy. As a result, rather than being a real forecast, the XGBoost outcomes are a conditional forecast. The time-of-day and day-of-week encodings as well as the lagged appliance-use features are variables that are truly known at the forecast origin; future indoor temperature and weather values would not be truly known, so the claimed XGBoost accuracy is probably optimistic in comparison to actual deployment.

6.6 Q6 Useful advice

For production implementation, XGBoost is advised. It was the most accurate model tested, with the lowest RMSE (67.06 on the same 24-hour window used for all other models, or 53.55 evaluated over the whole 14-day test period). In addition to accuracy, XGBoost offers feature importance rankings that make it comprehensible. Figure 8 illustrates the features that drive the forecast, such as hour-of-day and lagged appliance use, which aids in debugging and fostering trust. Given the conditional-forecast limitation covered in Q5, this recommendation is predicated on having access to generally credible weather forecasts for the exogenous covariates.

7. Future Improvements

- Fine-tuning Chronos-Bolt or comparable foundation models on this dataset specifically, rather than using it zero-shot.
- Extending SARIMAX to a SARIMAX-X specification with exogenous weather regressors, rather than relying on autoregressive structure alone.
- Using a proper rolling-origin (walk-forward) cross-validation scheme across many 24-hour windows, rather than a single held-out window, to obtain more robust performance estimates.
- Producing genuinely operational forecasts for the feature-based model by first forecasting the exogenous weather covariates themselves, rather than conditioning on their true future values.
- Exploring ensemble or hybrid approaches (e.g. SARIMAX residuals modelled by XGBoost) to combine the seasonal structure captured by SARIMAX with the nonlinear covariate interactions captured by XGBoost.

References

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